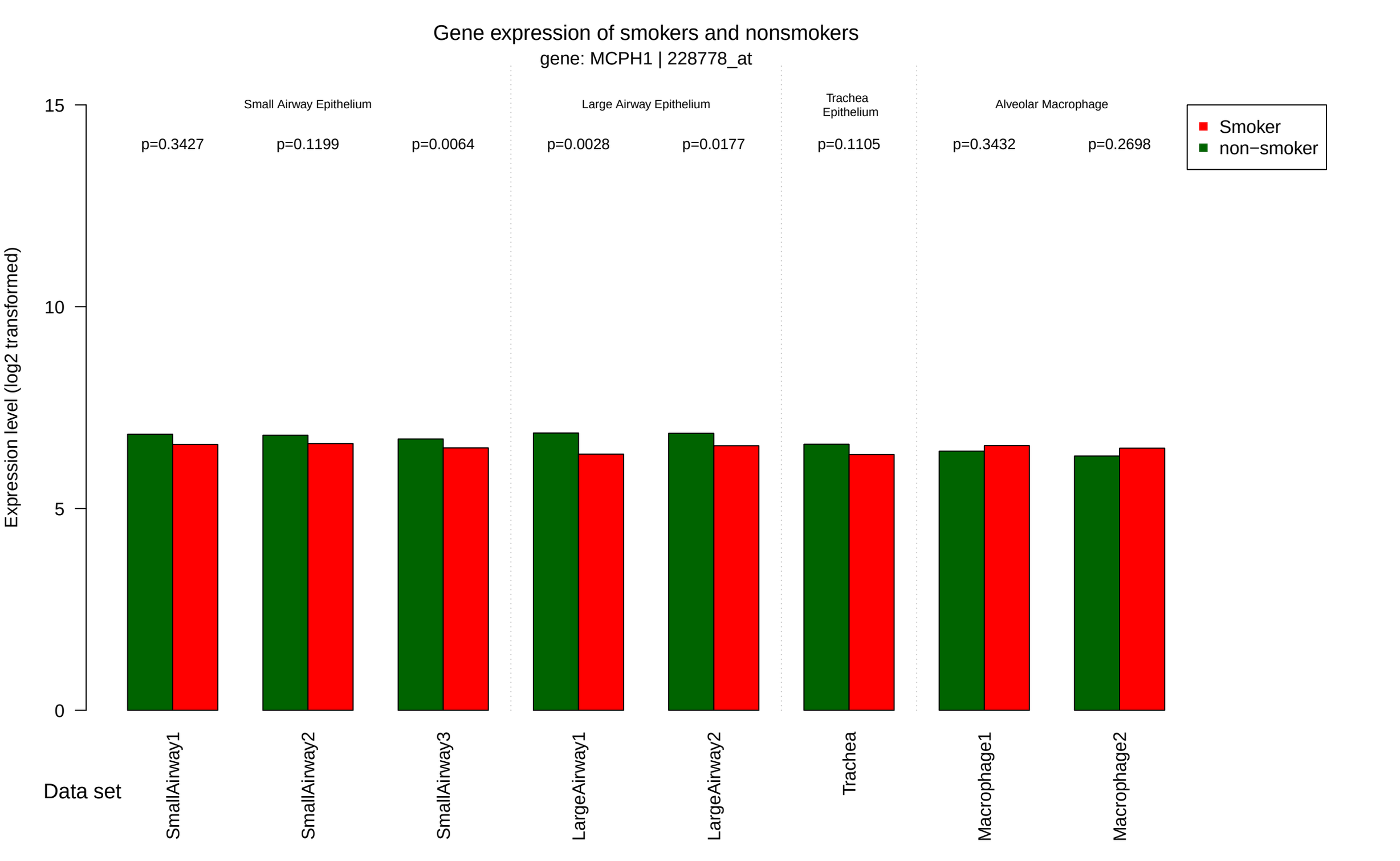
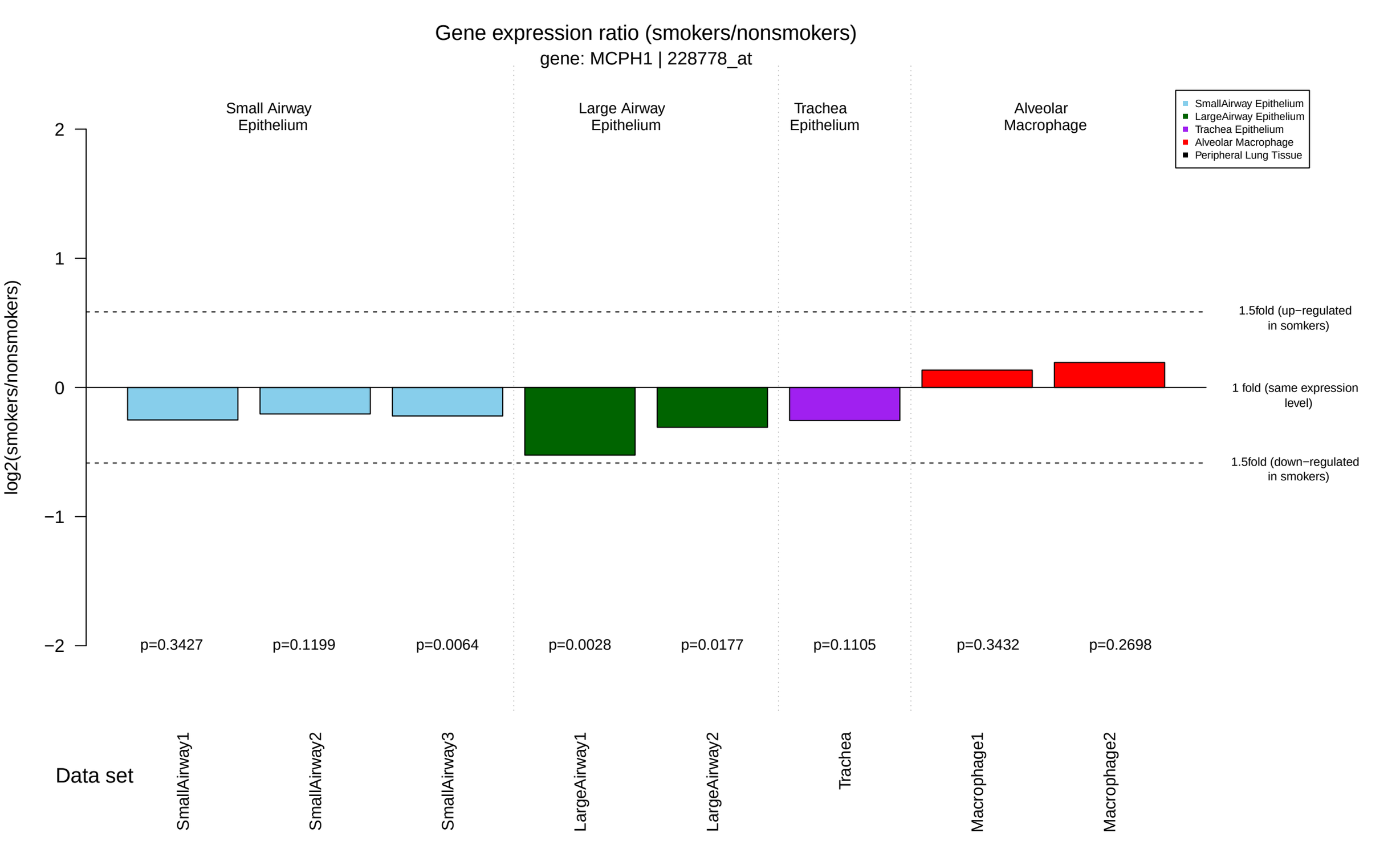
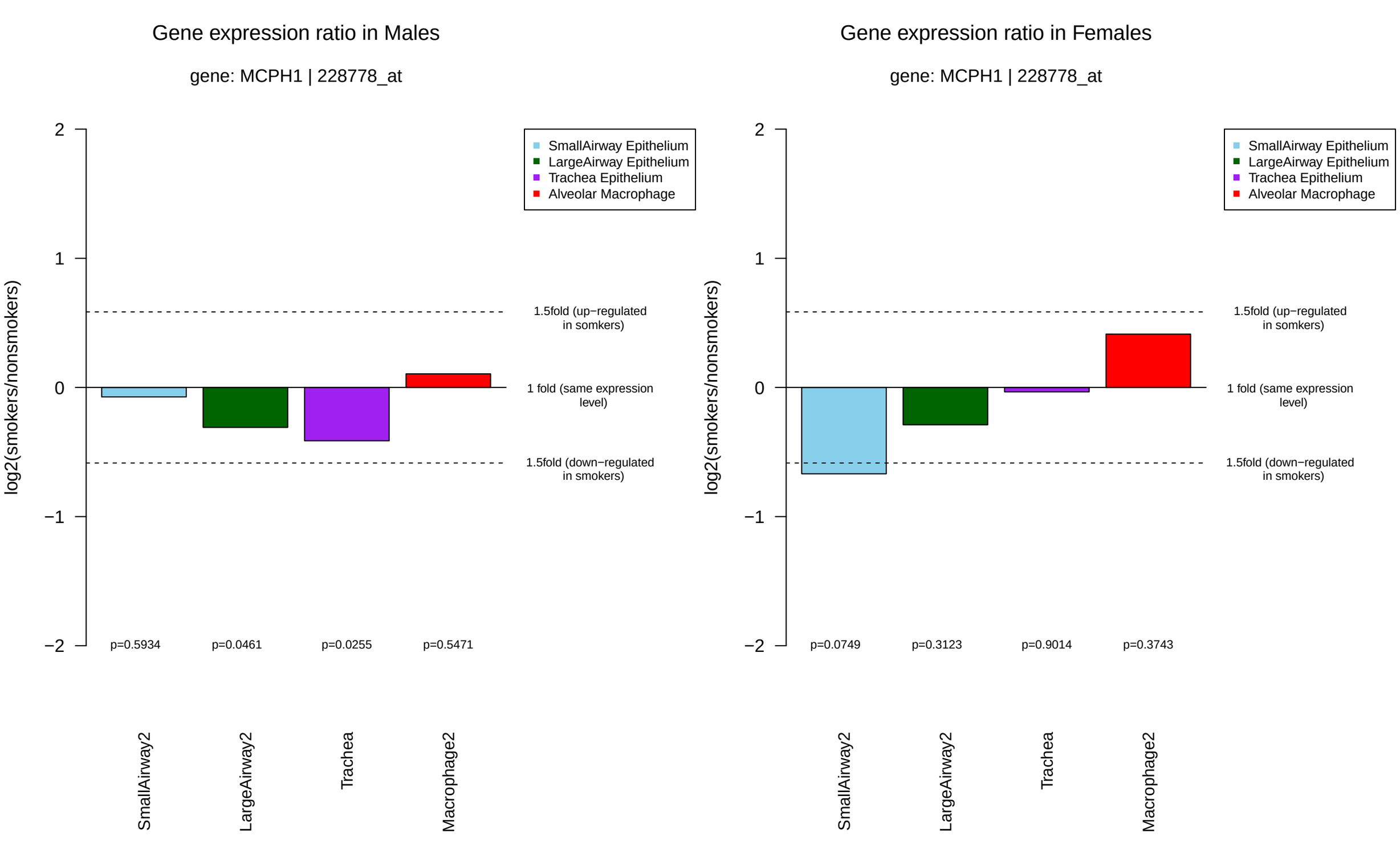
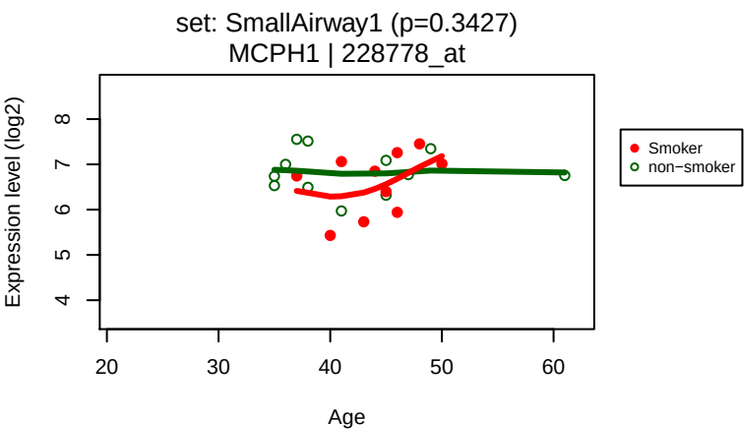
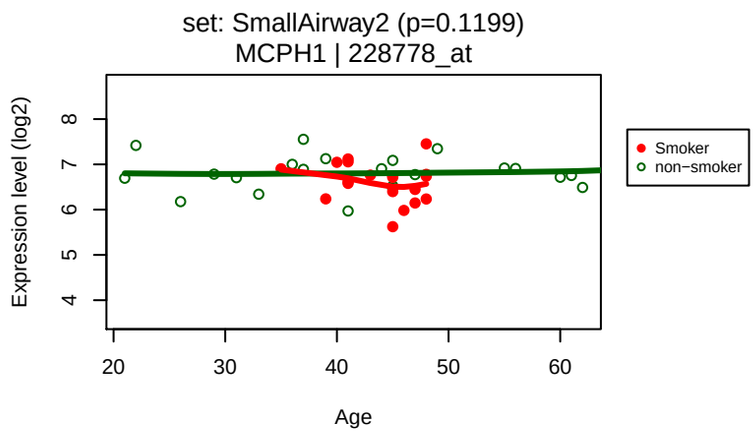
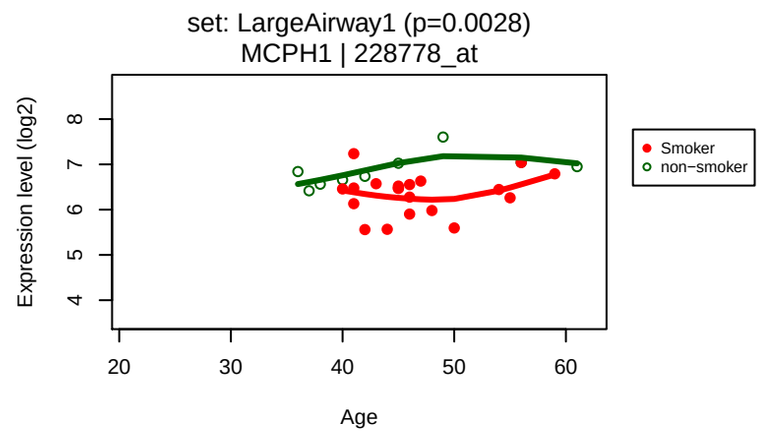

Gene expression of smokers and nonsmokers
gene: MCPH1 | 228778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3427
p=0.1199
p=0.0064
p=0.0028
p=0.0177
p=0.1105
p=0.3432
p=0.2698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

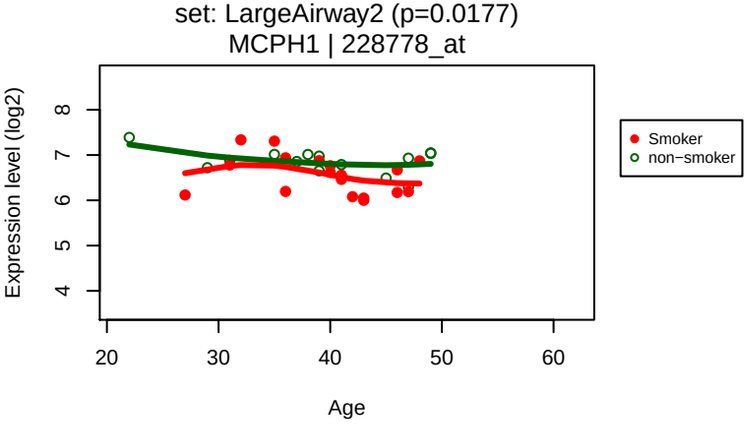

Gene expression ratio (smokers/nonsmokers)
gene: MCPH1 | 228778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3427
p=0.1199
p=0.0064
p=0.0028
p=0.0177
p=0.1105
p=0.3432
p=0.2698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

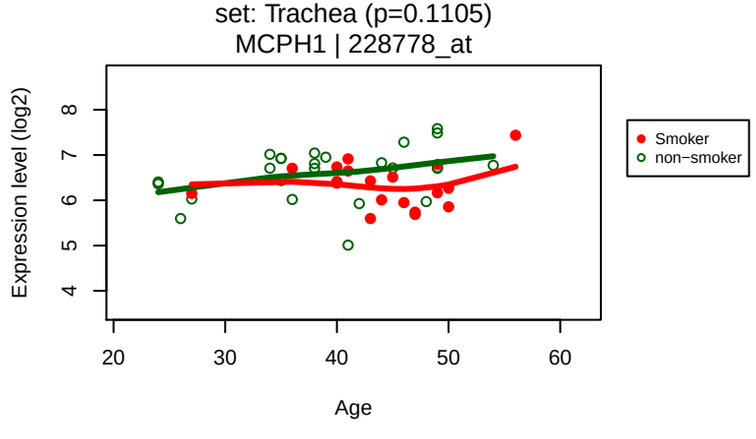

Gene expression ratio in Males
gene: MCPH1 | 228778_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5934
p=0.0461
p=0.0255
p=0.5471
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCPH1 | 228778_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0749
p=0.3123
p=0.9014
p=0.3743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

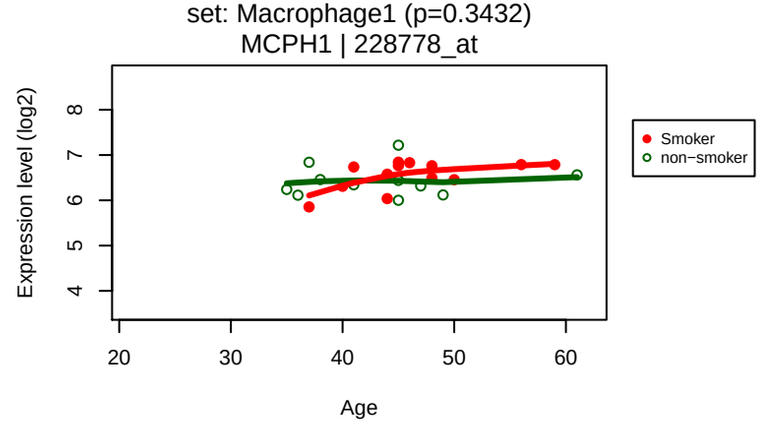

set: SmallAirway1 (p=0.3427)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker

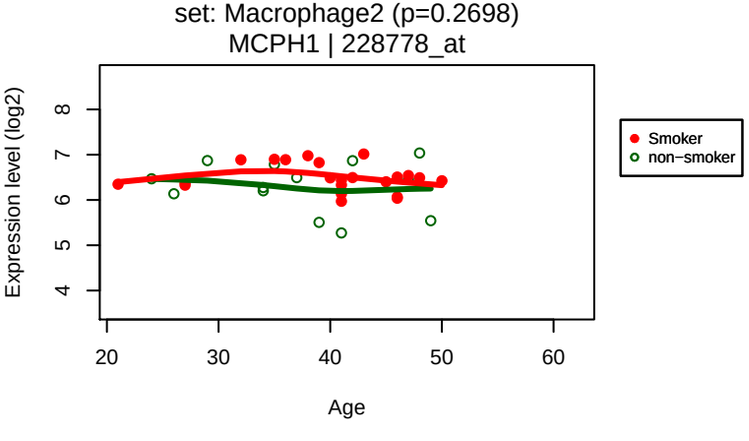

set: SmallAirway2 (p=0.1199)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker

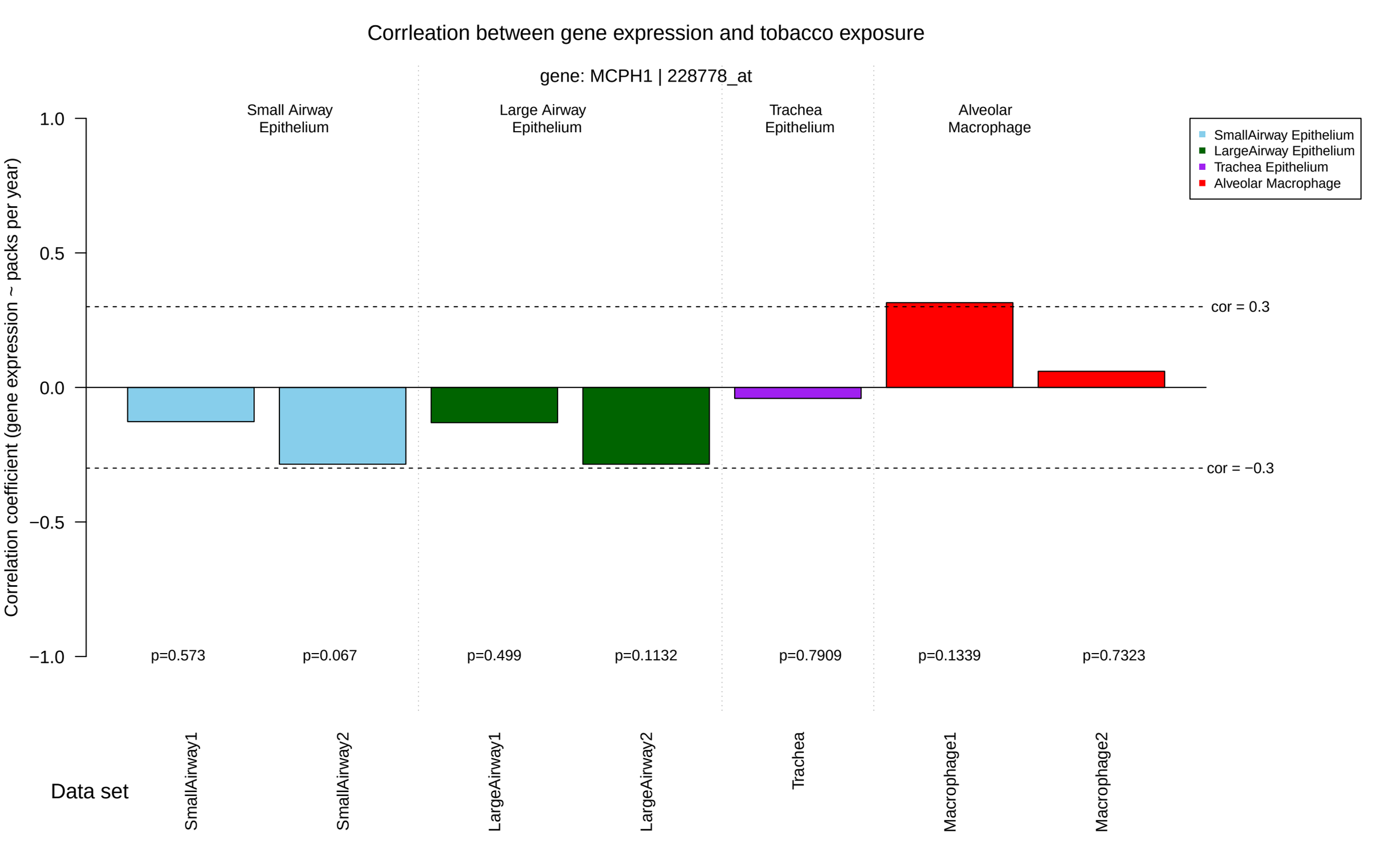

set: LargeAirway1 (p=0.0028)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0177)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1105)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3432)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2698)
MCPH1 | 228778_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCPH1 | 228778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.573
p=0.067
p=0.499
p=0.1132
p=0.7909
p=0.1339
p=0.7323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2